Supplementary Information
Why crop rankings do not determine land allocation

Author names and affiliations to be confirmed before submission

Stage II final scientific draft

1 Canonical theory and proof record

This section records the assumptions and arguments used in the main text. The versioned line-by-line proof package is in `theory/repaired/proofs.md`; theorem identifiers and transitions from the teacher Draft are in the canonical theorem registry. Historical numerical thresholds are not premises.

1.1 Objects and assumptions

Let $X \subset \mathbb{R}^n$ be the operational set in Eq. (2) of the main text. We assume X is non-empty and compact, $\widetilde{\pi}_i$ is integrable for every crop, and at least one $x \in X$ satisfies $r_\alpha(x) \leq \kappa$. A suitability vector s supplies only a weak order. It is not assumed to equal expected profit, a utility index or a Lagrangian derivative. The risk functional is loss-CVaR, so larger values are worse.

For a pair with $s_i > s_j$, define $h_{ij}(x) = x_j - x_i$ and tolerance $\tau_x \geq 0$. The possible and universal reversal indicators are generated by

$$g_{ij}^+ = \max_{x \in S_\kappa} h_{ij}(x), \qquad g_{ij}^- = \min_{x \in S_\kappa} h_{ij}(x). \tag{1}$$

Possible reversal is $g_{ij}^+ > \tau_x$; universal reversal is $g_{ij}^- > \tau_x$. A deterministic selector $\sigma(S_\kappa)$ defines selected reversal. This construction remains valid when the optimal set is a face rather than a point.

Theorem 1 (Existence and finite-scenario linearity)**.** *Under the assumptions above, the loss-CVaR problem has an optimizer and a convex feasible set. Under a finite distribution with positive scenario weights, it has an auxiliary linear-programme representation.*

Proof. Loss-CVaR is finite and continuous on the compact operational set under integrability. Its sublevel set intersected with X is closed, convex and, by assumption, non-empty. Compactness and continuity of the linear objective give existence. For scenarios $s = 1, \ldots, m$, substitute

$$r_\alpha(x) = \min_{v,q} \left\{ v + \frac{1}{1-\alpha} \sum_s w_s q_s : q_s \geq L_s(x) - v,\ q_s \geq 0 \right\}, \tag{2}$$

where v is free. The resulting objective and constraints are linear [Rockafellar and Uryasev, 2000, 2002]. $\square$

Proposition 1 (Ordinal insufficiency)**.** *A weak or strict suitability ranking alone does not identify a cardinal acreage allocation.*

Proof. The ranking retains only pairwise order relations. A strictly increasing transformation preserves all relations while changing cardinal score ratios. More directly, choose two non-empty compact feasible sets or two linear objectives that share the same external ranking but have different unique maximizers. The ranking is compatible with both allocations. Therefore a mapping, objective, feasible set and selection rule are required. □

Proposition 2 (Feasibility-forced reversal)**.** *For $s_i > s_j$, if $\sup_{x \in X} x_i + \tau_x < \inf_{x \in X} x_j$, every feasible allocation universally reverses the pair.*

Proof. The premise implies $x_i + \tau_x < x_j$ for all $x \in X$. Since $S_\kappa \subseteq X$, the inequality holds at every optimum. The simpler bound condition $u_i + \tau_x < \ell_j$ implies the premise. □

Proposition 3 (Slack-risk intersection)**.** *Let $S_0 = \arg\max_{x \in X} \mu^\top x$. If S_0 contains a risk-feasible point, the risk-constrained optimal value equals the unconstrained value and $S_\kappa = S_0 \cap \{x : r_\alpha(x) \leq \kappa\}$.*

Proof. A feasible member of S_0 attains the unconstrained upper bound in the constrained problem. Hence the values agree. Every constrained optimizer must attain that common value and therefore belongs to S_0; the reverse inclusion holds for risk-feasible members. One slack selected optimum does not imply that every member of S_0 is risk-feasible. □

Theorem 2 (Subgradient KKT characterization)**.** *Under a convex constraint qualification, or for a feasible bounded finite-scenario linear programme, a feasible x^* is optimal if and only if non-negative multipliers $\lambda, \gamma, \beta, \eta, a, b$ and $d \in \partial r_\alpha(x^*)$ satisfy stationarity and complementary slackness as stated in the main text.*

Proof. Minimize $-\mu^\top x$. The normal cone of the lower and upper bounds contributes $-a + b$; shared restrictions contribute $G^\top \eta$; land and budget contribute $\gamma \mathbf{1}$ and βc. Convex KKT conditions are necessary and sufficient. Pairwise subtraction cancels only γ. Under finite scenarios a valid CVaR subgradient is the negative scenario-profit average under tail weights whose total is one, whose upper bounds are $w_s/(1-\alpha)$ and whose atom weights may be fractional. □

Theorem 3 (Restricted dependence ordering)**.** *Fix marginals and a named copula family. If $L_{\theta_1}(x) \preceq_{\mathrm{cx}} L_{\theta_2}(x)$ for every allocation in a declared domain, loss-CVaR is weakly ordered there. If the domain is all of X, the more dispersed model has a subset of the risk-feasible policies and weakly lower optimal expected profit.*

Proof. CVaR is monotone under convex order for integrable losses. Pointwise risk ordering yields nested risk-feasible sets; maximizing the same linear objective over nested sets yields the value inequality. Crop-specific acreage need not be ordered because optimizers can move along different faces. □

Proposition 4 (Crossing-set inclusion)**.** *For any declared selector, the universal reversal region is contained in the selected region, which is contained in the possible region.*

Proof. The selected optimizer is an element of the optimal set. If the minimum gap exceeds tolerance, every element, including the selection, reverses. If the selected element reverses, the maximum gap exceeds tolerance. The argument implies no connectedness, monotonicity or uniqueness in the parameter indexing the sets. □

Theorem 4 (Information actionability)**.** *When ignoring a signal remains feasible, value of information is non-negative. A common posterior-optimal action gives zero value. Informed and uninformed optimized values are weakly non-decreasing under nested action sets, but their difference need not be monotone or strictly positive.*

Proof. The uninformed optimizer is an admissible constant contingent policy, so posterior optimization weakly improves the outer expectation. A common posterior optimizer makes the objectives equal. Set inclusion weakly raises each maximum separately, but the difference of two weakly increasing functions has no general monotonicity. E6 supplies constructive positive, zero and negative interactions under this theorem. □

1.2 Exact mechanism constructions

The four cases in main Fig. 2 use normalized two-crop acreage $x_1 + x_2 = 1$. In the margin construction, an external ranking $s_1 > s_2$ coexists with $\mu_2 > \mu_1$, so the profit optimum lies in $x_2 > x_1$. In the operational construction, bounds place all feasible points beyond the diagonal. In the risk construction, an admissible loss-CVaR half-space removes the unconstrained vertex and exposes a reversing boundary point. In the set-valued construction, the objective is constant along a feasible face crossing the diagonal. That face contains both reversal statuses, separating possible from universal reversal and making the selected status rule-dependent.

1.3 Teacher-Draft transitions

The original ranking-reversal equivalence is replaced by the KKT identity plus explicit displacement and feasibility certificates. Scalar tail monotonicity is replaced by restricted convex ordering. A unique threshold is replaced by possible and universal crossing sets. Pseudo-diversification remains a preregistered diagnostic without welfare content. Strict information–flexibility complementarity becomes a conditional hypothesis; the general theorem supports only non-negative information value when ignoring the signal is feasible.

2 Confirmatory simulation protocol and complete evidence ledger

2.1 Design and precision rule

The frozen E1–E6 design specifies factors, estimands, multiplicity, sequential checks, precision targets and maximum replication counts. An experiment meets the criterion only when every registered primary interval passes. E2 and E6 stop with precision achieved at $n = 16$ and $n = 16$, respectively. E1, E3, E4 and E5 reach $n = 64$ without meeting the complete experiment-level criterion. Their signs and isolated passing contrasts are not promoted.

2.2 E2 operational-intervention detail

E2 holds the stochastic margin law fixed and changes only the feasible set. The baseline is compared with a Soy contract, a rotation rule, their combination, a budget limit, budget–contract and budget–rotation combinations, the three-way combination, and a Corn-area cap. For every intervention we solve the registered deterministic selector and two auxiliary programmes that minimize and maximize each crop share over the full optimal face. This separates a solver-selected optimum from conclusions that are universal across all optima.

The selected Corn share falls from 1.000 at baseline to 0.318 under a budget constraint and 0.450 under the Corn cap. The registered universal-reversal contrasts are reported in the native probability scale and are accompanied by face-width, active-set and KKT diagnostics. Budget interventions act through the capital shadow price, rotation through intertemporal land coupling, contracts through minimum-demand geometry and the Corn cap through an explicit upper-bound multiplier. The tail-risk multiplier remains zero in the registered E2 environment, so the intervention result is a feasibility mechanism rather than evidence for downside-risk aversion. All 24 registered intervals satisfy the

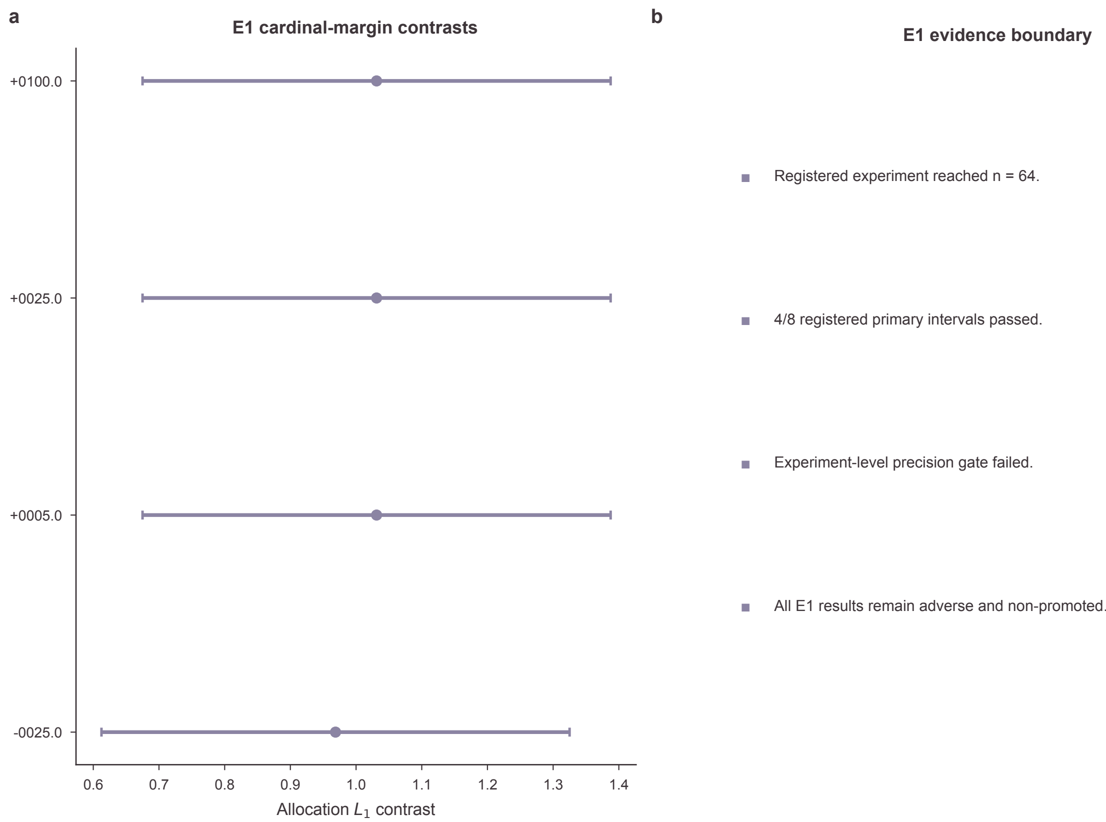

Figure S1: E1 cardinal-margin evidence. Registered allocation contrasts and the complete experiment-level decision are shown. E1 does not meet the pre-specified precision criterion at the replication ceiling.

experiment-level precision rule. Supplementary Fig. S7 expands these results into factorial response surfaces and estimand-level intervals.

2.3 E6 finite-state information environments

E6 makes decision timing explicit. Nature first draws a finite state; the planner observes the signal permitted by the registered information environment and then chooses an action from its corresponding measurable policy class. The restricted and enriched environments therefore have nested action sets, while the state probabilities, crop margins, constraints and risk functional remain fixed. Exact state enumeration avoids Monte Carlo ambiguity in the value comparison.

The three registered archetypes distinguish information that changes no action, information that substitutes for a robust option, and information that unlocks specialization. Their registered interaction estimates are 0.000, -3.114 (95% interval -3.476 to -2.751) and 3.258 (2.788 to 3.728), respectively. The result is not that more data always change acreage: value is weakly non-negative under nested action sets, but acreage changes only when the signal crosses a decision boundary. All three registered intervals satisfy the complete precision criterion.

2.4 E1 cardinal-margin experiment

E1 tests margin-driven reversal around registered gap contrasts. Its allocation contrasts are stable in sign, but the experiment-level criterion fails at the ceiling. Figure S1 preserves the interval geometry and the decision ledger.

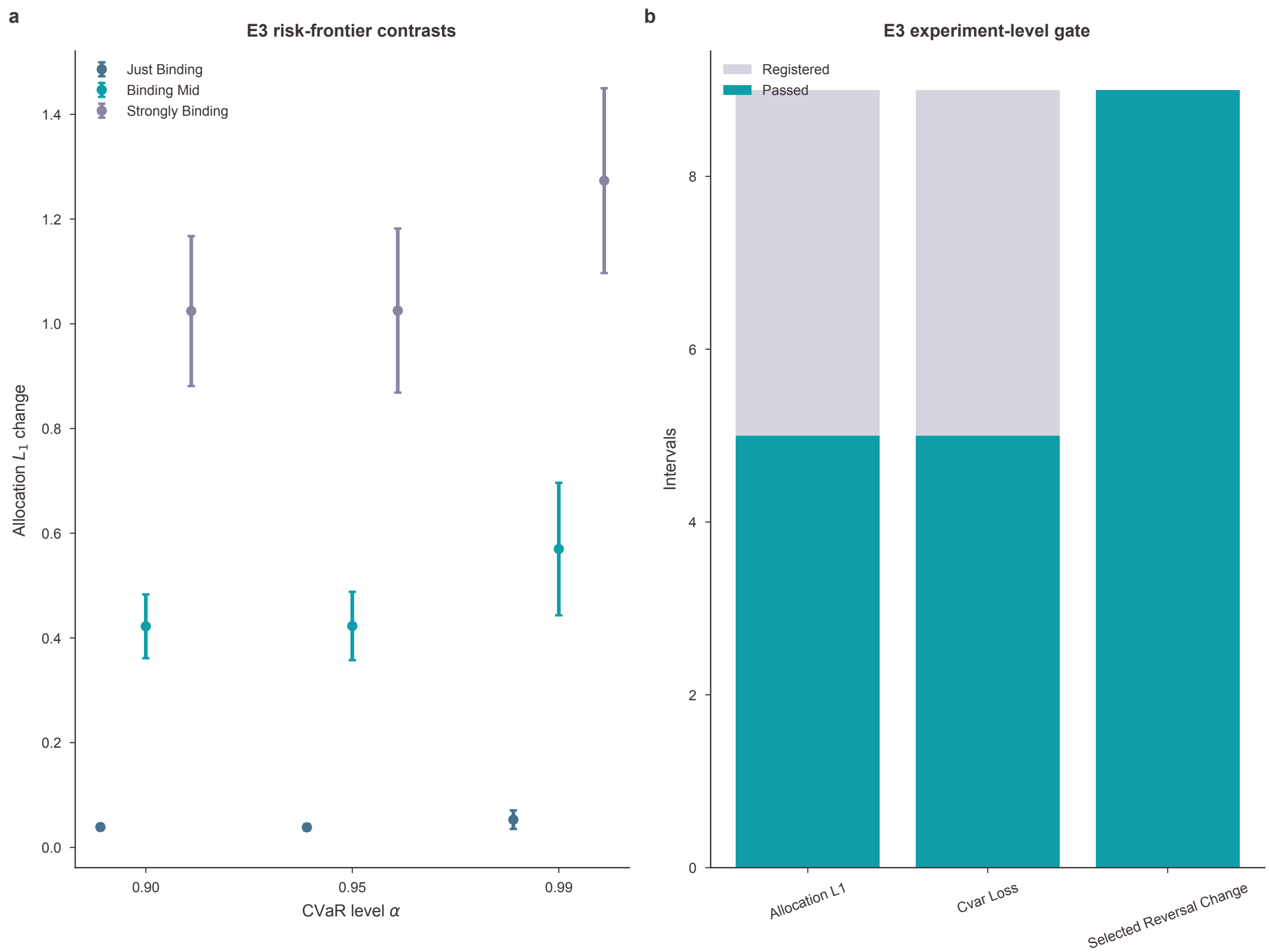

Figure S2: E3 risk-frontier evidence. Allocation contrasts are shown by CVaR level and binding regime beside the complete precision inventory. E3 remains inconclusive at the experiment level.

2.5 E3 downside-risk experiment

E3 varies loss-CVaR levels and binding regimes. Some individual allocation intervals are narrow, but the registered family contains intervals that do not meet their targets. All designed infeasible frontier cells remain recorded.

2.6 E4 and E5 dependence experiments

E4 tests dependence intensity within named families; E5 evaluates policies under paired true and assumed laws. Neither experiment meets its complete precision criterion. The cross-law pattern is retained as a diagnostic and cannot identify a universal dependence threshold or misspecification penalty.

2.7 Numerical integrity and attribution order

Reverse-order replay passes 12/12 checks and solver-method sensitivity passes 9/9. The maximum KKT stationarity residual is 1.82×10^{-11} and the maximum Shapley efficiency residual is 1.42×10^{-14}. The complete record includes all 206 infeasible rows. Order-path ranges expose the dependence of sequential

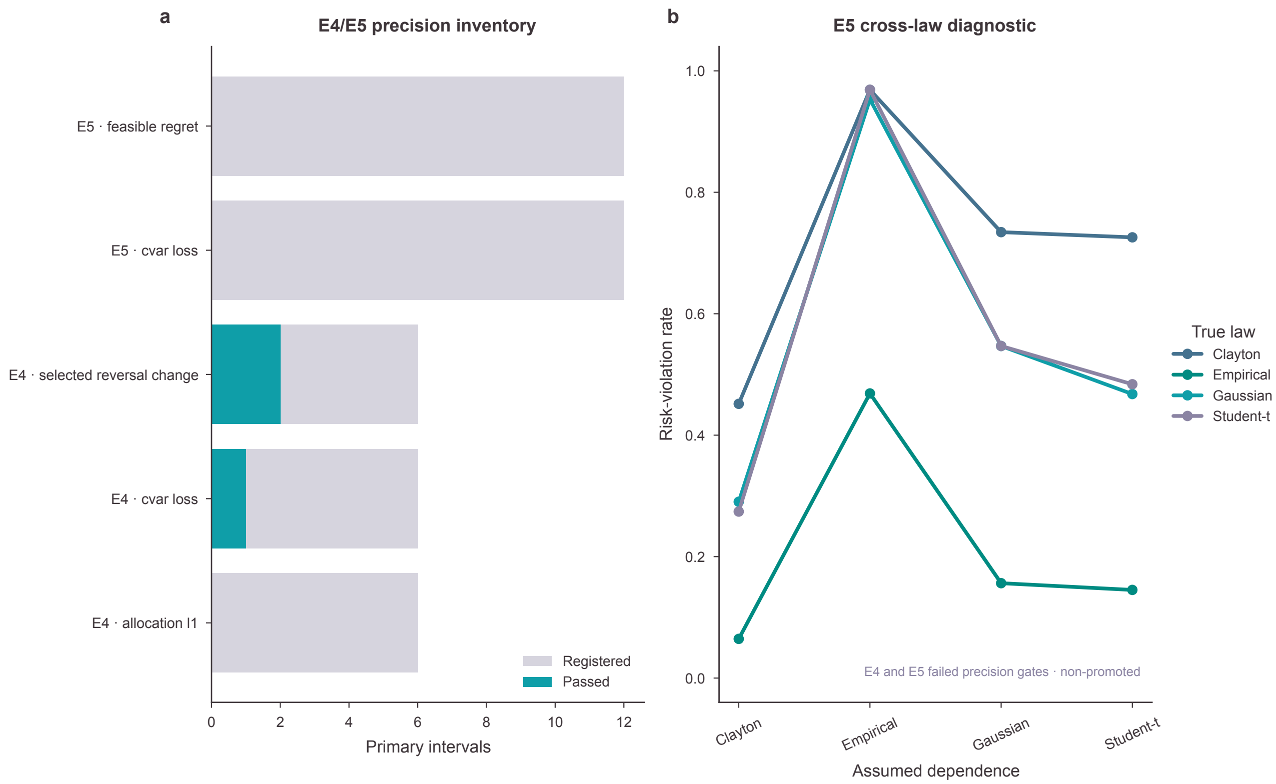

Figure S3: E4/E5 dependence evidence. The full precision inventory and E5 cross-law risk-violation diagnostic are shown. Both experiments remain inconclusive.

attribution on block order; the all-subset Shapley value is an accounting average rather than a causal decomposition.

3 Official-data extension and empirical robustness

3.1 Source reconstruction

USDA NASS Crop Production annual summaries are parsed in three non-overlapping blocks covering 2016–2024. The 2024 report reproduces the earlier frozen table exactly on state, year, crop, planted acreage and yield. USDA ERS crop-year prices and costs and the registered BLS CPI-U vintage provide national accounting inputs. Raw reports, URLs, retrieval timestamps and SHA-256 values are retained.

The complete-case rule yields 744 crop rows, 248 state-years and 31 states. A state-year enters only when corn, soybeans and winter wheat all have published planted acreage and yield. Missing and suppressed values are not imputed. The next state year was audited but excluded because the matching complete annual CPI-U construction was unavailable at the design freeze. County extraction was audited but not executed because no targeted credential-free official query was available; no county statement is made.

The parser reads 1035 crop records, of which 971 contain both acreage and yield. Requiring a complete three-crop state-year reduces the panel to 744 crop observations in 248 state-years. Coverage declines from 31 states in the early block to 26 in the middle block and 25 in the final block; the raw reports contain 1 missing acreage value and 64 missing yield values. These losses are displayed rather than repaired, and every retained map location is supported by the registered Census geometry.

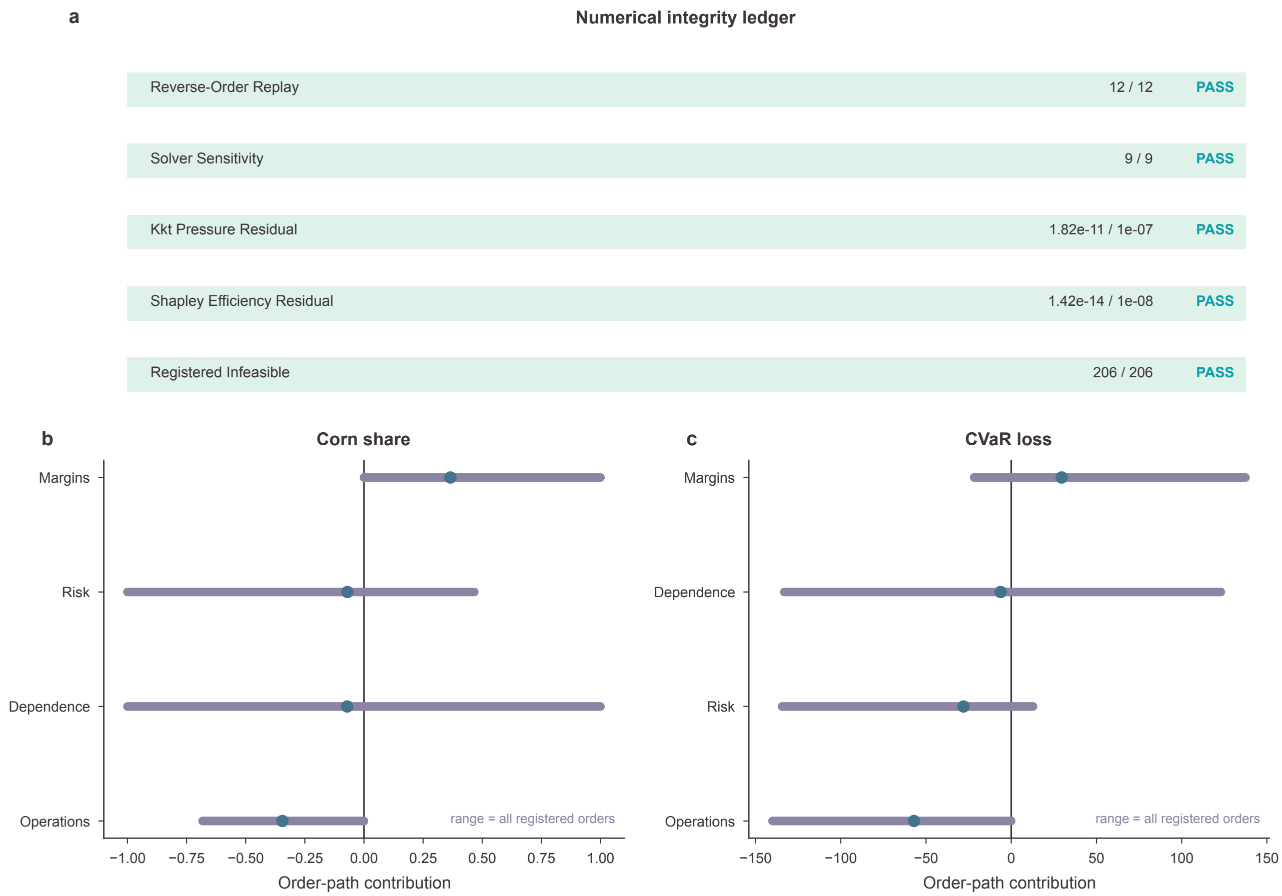

Figure S4: Numerical integrity and attribution-order sensitivity. Replay, solver, KKT, Shapley and infeasibility checks are reported with all registered block-order ranges.

3.2 Definitions and estimands

Relative yield normalizes each state yield by its same-year national value. Standardized revenue multiplies state yield by national price. Operating- and total-cost margins subtract the corresponding national cost after real-dollar conversion. These scores are transparent accounting constructions, not local realized margins.

Concurrent metrics retain ties and include Kendall's tau-b, Spearman correlation when defined, inversion intensity and top-rank disagreement. The primary temporal model uses prior score-leader status and prior acreage share with crop, year and state fixed effects. All scores precede acreage-share change. Uncertainty uses 5000 state-cluster draws.

3.3 Robustness and identification

Annual summaries show that definition sensitivity persists through the expanded period. Leave-one-state-out ranges show that no single retained state creates the aggregate pattern. The sample-flow display exposes every complete-case restriction. National relative yield is tied in all 9 years, illustrating a definitional aggregation boundary.

For operating margin, leave-one-state-out mean inversion ranges from 0.399 to 0.421. The temporal design contains 2604 crop-level transitions and 868 strictly ordered state-year transitions for each score definition. These estimates condition on prior acreage share and fixed effects, but they remain

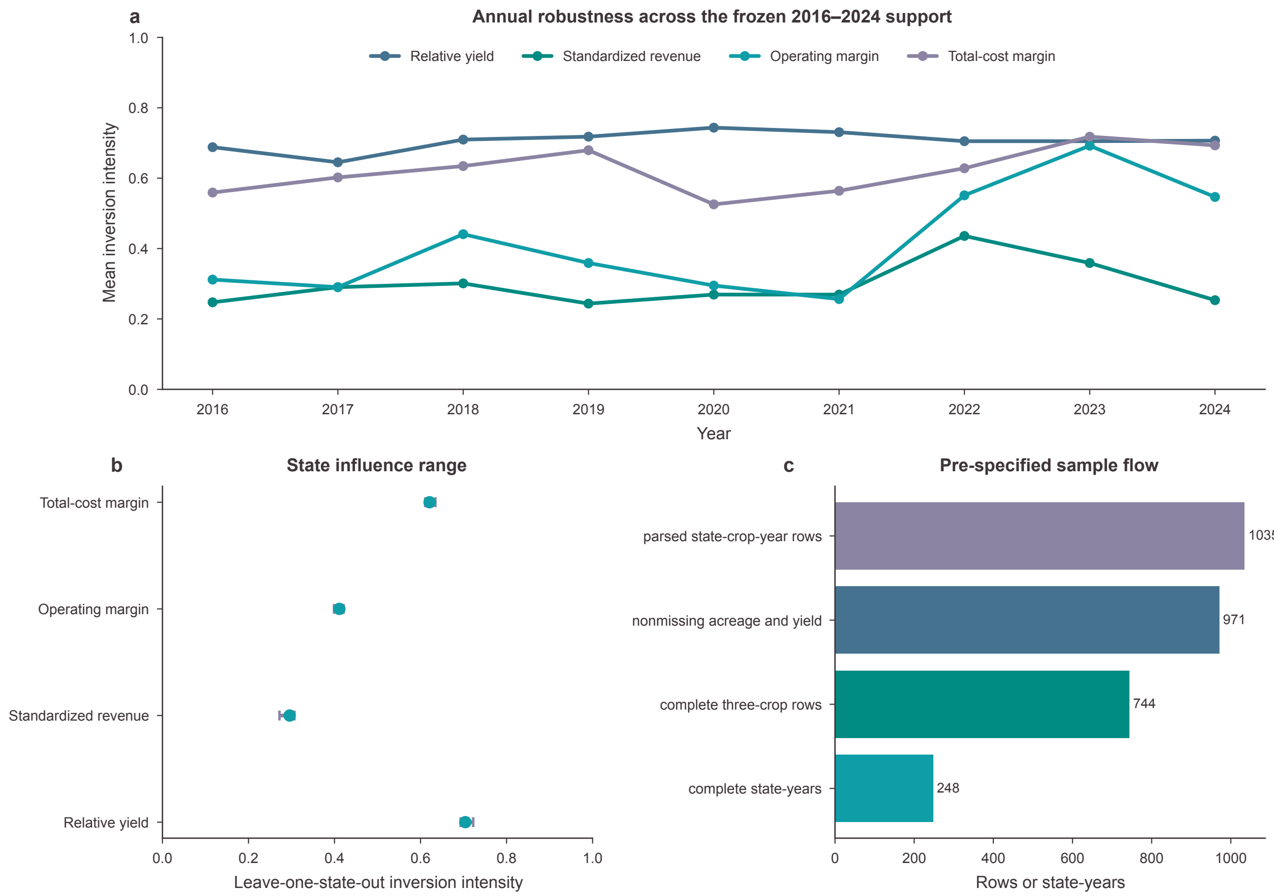

Figure S5: Empirical robustness and sample construction. Annual inversion series, leave-one-state-out ranges and the complete pre-specified sample flow are shown for the 2016–2024 official-data panel.

descriptive because both observed scores and acreage may respond to unmeasured local constraints, expectations and policy exposure.

The data identify descriptive score–acreage relations and strictly ordered transitions. They do not identify observed acreage as optimal; private land, budget, rotation or contract restrictions; farmer beliefs or objective weights; a binding downside-risk limit; a dependence mechanism; causal responses; welfare; or regret relative to an unobserved optimum.

4 Reproducibility, lineage and artifact inventory

The repository is organized as an evidence graph. Immutable raw sources are checksum-registered; processing produces canonical panels; theory, simulation and empirical modules write tidy outputs; visualization reads only verified outputs; and manuscript macros read named fields from those outputs. The path for each major statement is claim → figure or table → source data → code → configuration → execution record.

4.1 Empirical reproducibility

The GOAL-16 empirical command starts from checksum-registered official reports and writes tidy panels, estimands, bootstrap draws, source records, lineage and validation metadata. Two isolated executions reproduce every artifact byte for byte. The validator checks 744 crop rows, 248 state-years,

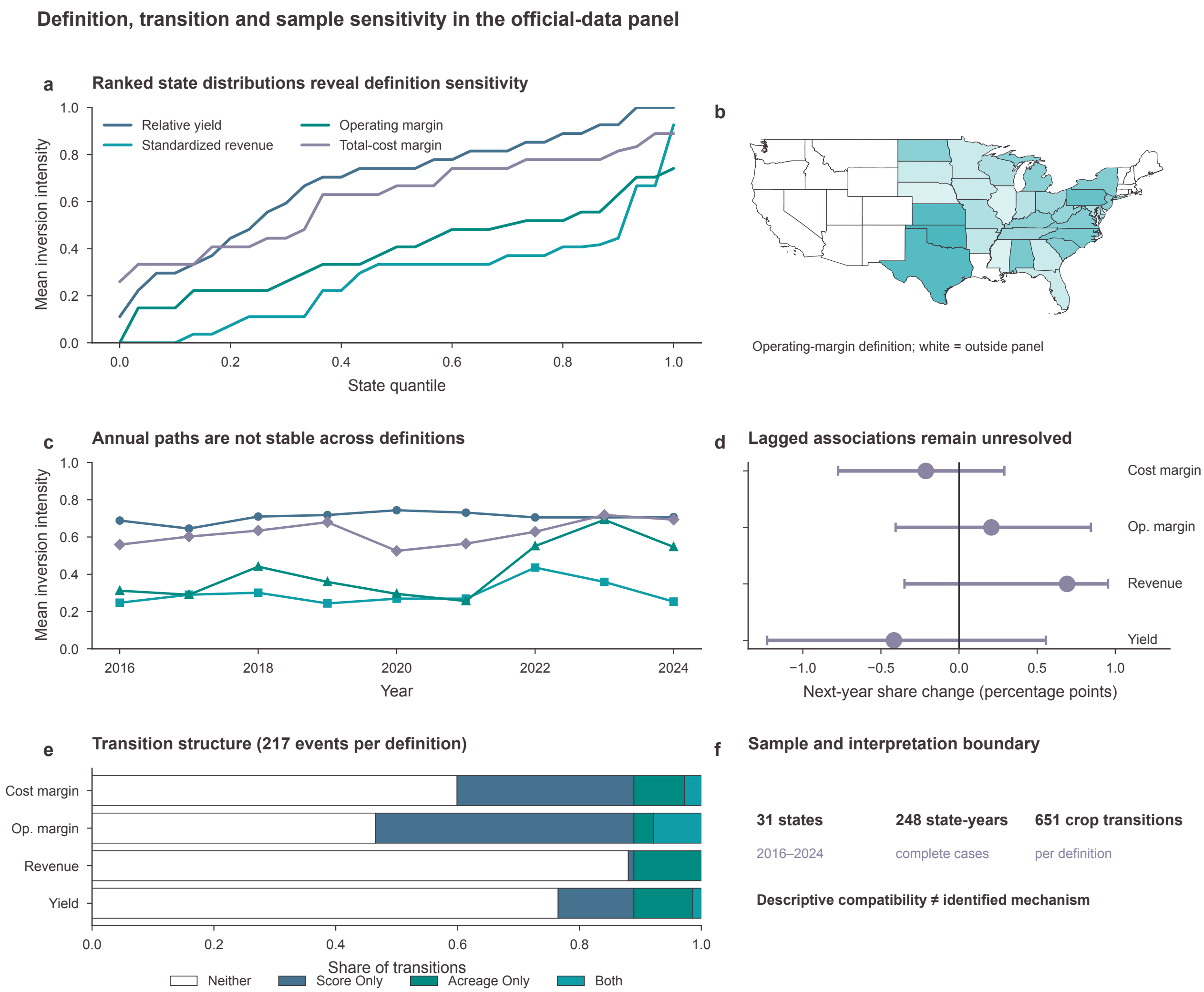

Figure S6: Expanded empirical diagnostics. Ranked score–acreage distributions, yearly inversion paths, lagged leader estimates in percentage points, transition counts and the fixed-vintage state coverage map expose the magnitude, time structure and support of the official-data analysis. White map units are outside the complete-case panel and are not interpreted as zero.

31 states, 2604 crop-transition rows, 868 rank-transition events, 5000 bootstrap draws, score definitions, timing and aggregation ties.

4.2 Simulation reproducibility

The confirmatory command resolves the frozen design hash, scenario seeds and task registry before solving. Output validators reconcile raw, contrast, face, pressure, information, attribution and stopping ledgers. Independent replay reverses task order; solver sensitivity reruns named anchors. The validator fails closed if any unexplained infeasibility, missing adverse row, promotion-rule violation, KKT excess or attribution non-closure appears.

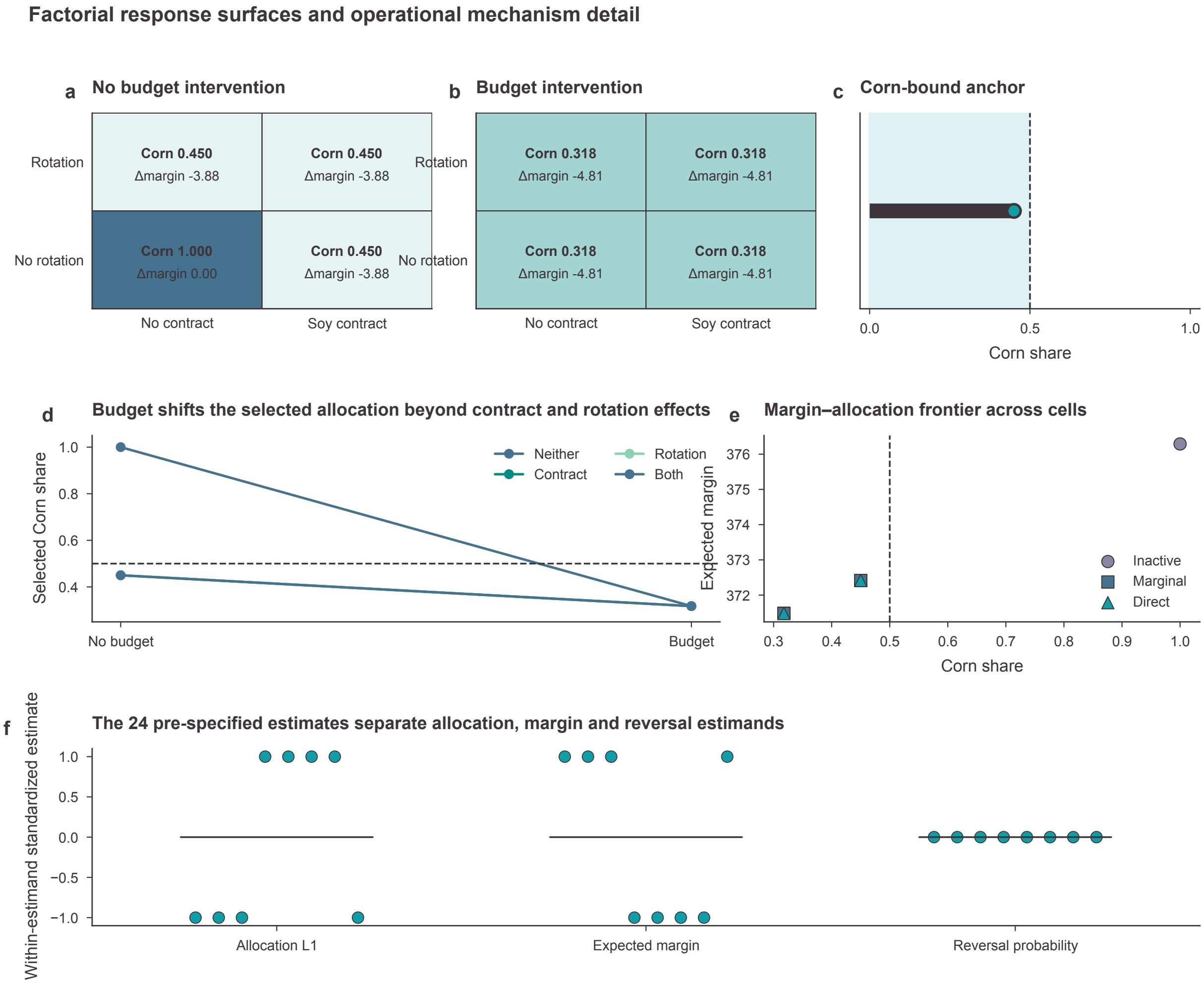

Figure S7: E2 factorial response surfaces and mechanism diagnostics. The supplement expands the main-text intervention matrix into Corn-share response surfaces, the Corn-bound anchor, the budget slope, the margin–allocation frontier and estimand-level intervals. Thresholds, face classification and KKT channels are shown in aligned panels so that visual proximity is not mistaken for a shared mechanism.

4.3 Figure production and colour contract

The figure build exports editable SVG and PDF, 300-dpi PNG and 600-dpi TIFF for six GOAL-17 main figures, two new GOAL-17 supplementary figures and the five preceding confirmatory supplementary figures. Each new figure has tidy source data, a caption, a manifest row and checksums. Exact dimensions are frozen at the 183-mm final width. The only base colours are `#3D3539`, `#0F9EA8`, `#008B82`, `#45728F`, `#8CD1B2` and `#8B84A3`; transparent derivatives are permitted only where overlapping distributions or map layers require them. Two materially different final-size candidates were compared for every main figure before selection. Full-size, grayscale, deuteranopia and protanopia proofs are generated and inspected. Renderer-bounds and title-collision validators report zero failures, and every selected composition reserves separate lanes for prose, labels and plotted marks.

4.4 Manuscript and release build

The main source is `manuscript/main.tex`; this document is `supplementary/supplementary.tex`. The manuscript-number generator records every displayed macro, unit, output file and field in a CSV registry. Claim–citation and figure-use registries restrict citations to verified literature and scientific statements to admitted evidence levels. All 44 teacher-Draft content rows retain a final disposition; unsupported historical numbers and universal mechanism claims remain excluded.

The release build compiles both documents twice in isolated directories with a fixed source date and timezone. It rejects LaTeX errors, undefined citations or references, missing assets and overfull boxes. Byte-identical PDFs are copied to the release directory, every page is rendered for visual review, font embedding is checked, and package checksums are regenerated. The final archive includes manuscript sources, Supplementary Information, six main and seven supplementary figures in four formats, source data, registries, audit reports and reproduction instructions.

Software environment. Python 3.11 is managed by the locked project environment. Optimization uses SciPy/HiGHS; data processing uses pandas and NumPy; figures use Matplotlib under the frozen Stage II style contract. LaTeX uses pdfLaTeX, BibTeX and `latexmk`; Poppler renders page proofs. Exact tool versions are written to the build-environment record.

Author-owned metadata. Author names, affiliations, ORCID identifiers, funding, contributions, acknowledgements and the final competing-interests statement remain explicit placeholders. Their absence does not alter scientific results, but the package is a final scientific draft rather than a journal-submission archive until authors complete them.

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R. Tyrrell Rockafellar and Stanislav Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, 2002. doi: 10.1016/S0378-4266(02)00271-6. URL `https://doi.org/10.1016/S0378-4266(02)00271-6`.